

Processando config 7, seed 4, teste: Rabbit
 xx
 Starting Rabbit: nb = 16777216
 Version: TestU01 1.2.3
 xx

 HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
 Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config7_seed4.bin

smultin_MultinomialBitsOver test:

 N = 2, n = 8388608, r = 0, s = 32, L = 38,
 Sparse = TRUE

 Number of bits = n = 8388608
 Number of cells = 2^L = 274877906944
 Expected number per cell = $1 / 32768$
 EColl = $n^2 / (2k)$ = 128
 Hashing = TRUE

 Collision test
 CollisionOver: density = $n / k = 1 / 32768.00$
 Expected number of collisions = $\mu = 128.00$

 Results of CollisionOver test:

POISSON approximation	:	
Expected number of collisions = $N \cdot \mu$	:	256.00
Observed number of collisions	:	277
p-value of test	:	0.10

 Total number of cells containing j balls

j = 0	:	549739036949
j = 1	:	16776662
j = 2	:	277
j = 3	:	0
j = 4	:	0
j = 5	:	0

 CPU time used : 00:00:04.44

Generator state:
 16777216 bits have been read.

 HOST = matheus-IdeaPad-3-15ALC6, Linux

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snpair_ClosePairsBitMatch test:

N = 1, n = 262144, r = 0, t = 2

Max of all bit distances : 17.00
p-value of test : 0.39

CPU time used : 00:00:00.05

Generator state:
33554432 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
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snpair_ClosePairsBitMatch test:

N = 1, n = 131072, r = 0, t = 4

Max of all bit distances : 8.00
p-value of test : 0.12

CPU time used : 00:00:00.08

Generator state:
50331648 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config7_seed4.bin

svaria_AppearanceSpacings test:

N = 1, Q = 524288, K = 524288, r = 0, s = 30, L = 15

Sequences of $n = (K + Q)L = 15728640$ bits
Q = 524288 initialization blocks
K = 524288 blocks for the test
the blocks have L = 15 bits

Normal statistic : -0.34
p-value of test : 0.63

CPU time used : 00:00:00.00

Generator state:
67108864 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config7_seed4.bin

scomp_LinearComp test:

N = 1, n = 18176, r = 0, s = 32

Number of degrees of freedom : 7
Chi2 statistic for size of jumps : 8.03
p-value of test : 0.33

Normal statistic for number of jumps : 0.31
p-value of test : 0.38

CPU time used : 00:00:00.18

Generator state:
67127040 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config7_seed4.bin

scomp_LempelZiv test:

N = 1, n = 16777216, r = 0, s = 32, k = 24

Normal statistic : 1.71
p-value of test : 0.04

CPU time used : 00:00:00.40

Generator state:

83904256 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config7_seed4.bin

sspectral_Fourier1 test:

N = 1, n = 1048576, r = 0, s = 32, k = 20

Normal statistic : -0.099
p-value of test : 0.54

CPU time used : 00:00:00.21

Generator state:

84952832 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config7_seed4.bin

sspectral_Fourier3 test:

N = 8065, n = 2048, r = 0, s = 32, k = 11

Kolmogorov-Smirnov+ statistic = D+ : 0.040
p-value of test : 0.18

Kolmogorov-Smirnov- statistic = D- : 0.040
p-value of test : 0.19

Anderson-Darling statistic = A2 : 1.91
p-value of test : 0.10

CPU time used : 00:00:00.12

Generator state:

101728032 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config7_seed4.bin

sstring_LongestHeadRun test:

N = 1, n = 60, r = 0, s = 32, L = 279616

Number of degrees of freedom : 3
Chi-square statistic : 1.84
p-value of test : 0.61

Global longest run of 1 : 23.00
p-value of test : 0.50

CPU time used : 00:00:00.06

Generator state:
118504992 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config7_seed4.bin

sstring_PeriodsInStrings test:

N = 1, n = 524288, r = 0, s = 31

Number of degrees of freedom : 94
Chi-square statistic : 108.99
p-value of test : 0.14

CPU time used : 00:00:00.02

Generator state:
135282208 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config7_seed4.bin

sstring_HammingWeight test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Number of degrees of freedom : 22
Chi-square statistic : 14.46
p-value of test : 0.88

CPU time used : 00:00:00.02

Generator state:
152059424 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config7_seed4.bin

sstring_HammingCorr test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Normal statistic : -1.13
p-value of test : 0.87

CPU time used : 00:00:00.00

Generator state:
168836640 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config7_seed4.bin

sstring_HammingCorr test:

N = 1, n = 262144, r = 0, s = 32, L = 64

Normal statistic : 0.72
p-value of test : 0.24

CPU time used : 00:00:00.00

Generator state:
185613856 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config7_seed4.bin

sstring_HammingCorr test:

N = 1, n = 131072, r = 0, s = 32, L = 128

Normal statistic : -0.69
p-value of test : 0.76

CPU time used : 00:00:00.00

Generator state:
202391072 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config7_seed4.bin

sstring_HammingIndep test:

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 177
Chi-square statistic : 167.46
p-value of test : 0.68

CPU time used : 00:00:00.02

Generator state:
219168288 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config7_seed4.bin

sstring_HammingIndep test:

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 305
Chi-square statistic : 303.40
p-value of test : 0.52

CPU time used : 00:00:00.00

Generator state:

235945504 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config7_seed4.bin

sstring_HammingIndep test:

N = 1, n = 131072, r = 0, s = 32, L = 64, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 489
Chi-square statistic : 457.63
p-value of test : 0.84

CPU time used : 00:00:00.00

Generator state:

252722720 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config7_seed4.bin

sstring_AutoCor test:

N = 1, n = 16777153, r = 0, s = 32, d = 1

```
-----  
Normal statistic          :   -1.35  
p-value of test          :    0.91  
-----
```

```
CPU time used             :   00:00:00.02
```

```
Generator state:  
269499904 bits have been read.
```

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*****
```

```
HOST = matheus-IdeaPad-3-15ALC6, Linux
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ufile_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config7_seed4.bin
```

```
sstring_AutoCor test:
```

```
-----  
N = 1, n = 16777154, r = 0, s = 32, d = 2  
-----
```

```
-----  
Normal statistic          :    0.69  
p-value of test          :    0.24  
-----
```

```
CPU time used             :   00:00:00.00
```

```
Generator state:  
286277088 bits have been read.
```

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*****
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```
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config7_seed4.bin
```

```
sstring_Run test:
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```
-----  
N = 1, n = 3355443, r = 0, s = 32  
-----
```

```
-----  
Total number of 1 runs: 3355449
```

```
Number of degrees of freedom : 36  
Chi2 statistic for number of runs : 23.21  
p-value of test : 0.95  
-----
```

```
Total number of bits: 13424896
```

```
Normal statistic for number of bits : 0.60  
p-value of test : 0.27
```

CPU time used : 00:00:00.05

Generator state:
299701984 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config7_seed4.bin

smarsa_MatrixRank test:

N = 1, n = 16384, r = 0, s = 32, L = 32, k = 32

Number of degrees of freedom : 3
Chi-square statistic : 4.18
p-value of test : 0.24

CPU time used : 00:00:00.06

Generator state:
316479200 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config7_seed4.bin

smarsa_MatrixRank test:

N = 1, n = 163, r = 0, s = 32, L = 320, k = 320

Number of degrees of freedom : 2
Chi-square statistic : 1056.73
p-value of test : eps *****

CPU time used : 00:00:00.09

Generator state:
333170400 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config7_seed4.bin

swalk_RandomWalk1 test:

N = 1, n = 131072, r = 0, s = 32, L0 = 128, L1 = 128

Test on the values of the Statistic H

Number of degrees of freedom	:	42
ChiSquare statistic	:	46.52
p-value of test	:	0.29

Test on the values of the Statistic M

Number of degrees of freedom	:	42
ChiSquare statistic	:	43.21
p-value of test	:	0.42

Test on the values of the Statistic J

Number of degrees of freedom	:	64
ChiSquare statistic	:	87.88
p-value of test	:	0.03

Test on the values of the Statistic R

Number of degrees of freedom	:	36
ChiSquare statistic	:	55.72
p-value of test	:	0.02

Test on the values of the Statistic C

Number of degrees of freedom	:	22
ChiSquare statistic	:	32.79
p-value of test	:	0.06

CPU time used : 00:00:00.05

Generator state:
349947616 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config7_seed4.bin

swalk_RandomWalk1 test:

N = 1, n = 16384, r = 0, s = 32, L0 = 1024, L1 = 1024

Test on the values of the Statistic H

Number of degrees of freedom	:	94
ChiSquare statistic	:	97.00
p-value of test	:	0.40

Test on the values of the Statistic M

Number of degrees of freedom	:	94
ChiSquare statistic	:	93.20
p-value of test	:	0.50

Test on the values of the Statistic J

Number of degrees of freedom	:	512
ChiSquare statistic	:	504.85
p-value of test	:	0.58

Test on the values of the Statistic R

Number of degrees of freedom	:	90
ChiSquare statistic	:	85.78
p-value of test	:	0.61

Test on the values of the Statistic C

Number of degrees of freedom	:	50
ChiSquare statistic	:	40.26
p-value of test	:	0.84

CPU time used : 00:00:00.05

Generator state:

366724832 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config7_seed4.bin

swalk_RandomWalk1 test:

N = 1, n = 1675, r = 0, s = 32, L0 = 10016, L1 = 10016

Test on the values of the Statistic H

Number of degrees of freedom	:	130
ChiSquare statistic	:	137.09
p-value of test	:	0.32

Test on the values of the Statistic M

Number of degrees of freedom	:	131
ChiSquare statistic	:	120.59
p-value of test	:	0.73

Test on the values of the Statistic J

Number of degrees of freedom	:	161
ChiSquare statistic	:	130.84
p-value of test	:	0.96

Test on the values of the Statistic R

Number of degrees of freedom	:	130
ChiSquare statistic	:	136.33
p-value of test	:	0.33

Test on the values of the Statistic C

Number of degrees of freedom	:	89
ChiSquare statistic	:	60.64
p-value of test	:	0.9907

CPU time used : 00:00:00.03

Generator state:

383501632 bits have been read.

===== Summary results of Rabbit =====

Version:	TestU01 1.2.3
Generator:	ufile_CreateReadBin
Number of bits:	16777216
Number of statistics:	39

Total CPU time: 00:00:06.11
The following tests gave p-values outside [0.001, 0.9990]:
(eps means a value < 1.0e-300):
(eps1 means a value < 1.0e-15):

Test	p-value
-----	-----
22 MatrixRank, 320 x 320	eps
-----	-----

All other tests were passed